DeepWAF-XAI: Master Source Data & Literature Benchmark Inventory

Section 8.4 Graphical Analysis: Underlying literature benchmarks, measured project evaluation, and comparal

System Name	Architecture / Method Type	Metric Name
Traditional Rule-Based WAF (ModSecurity CRS)	Deterministic Regex Rule Matching	Accuracy
Traditional Rule-Based WAF (ModSecurity CRS)	Deterministic Regex Rule Matching	Precision
Traditional Rule-Based WAF (ModSecurity CRS)	Deterministic Regex Rule Matching	Recall
Traditional Rule-Based WAF (ModSecurity CRS)	Deterministic Regex Rule Matching	F1-Score
Classical ML Baseline (SVM / RF)	N-Gram TF-IDF + Random Forest / SVM	Accuracy
Classical ML Baseline (SVM / RF)	N-Gram TF-IDF + Random Forest / SVM	Precision
Classical ML Baseline (SVM / RF)	N-Gram TF-IDF + Random Forest / SVM	Recall
Classical ML Baseline (SVM / RF)	N-Gram TF-IDF + Random Forest / SVM	F1-Score
BALNet (CNN Base Paper, 2026)	1D Convolutional Neural Network + ADASYN	Accuracy
BALNet (CNN Base Paper, 2026)	1D Convolutional Neural Network + ADASYN	Precision
BALNet (CNN Base Paper, 2026)	1D Convolutional Neural Network + ADASYN	Recall
BALNet (CNN Base Paper, 2026)	1D Convolutional Neural Network + ADASYN	F1-Score
UniEmbed (Feature Fusion MLP, 2025)	Multi-Head Self-Attention + Feature Fusion MLP	Accuracy
UniEmbed (Feature Fusion MLP, 2025)	Multi-Head Self-Attention + Feature Fusion MLP	Precision
UniEmbed (Feature Fusion MLP, 2025)	Multi-Head Self-Attention + Feature Fusion MLP	Recall

UniEmbed (Feature Fusion MLP, 2025)	Multi-Head Self-Attention + Feature Fusion MLP	F1-Score
HawkPhish-DNN (HHO-DNN, 2025)	Harris Hawks Optimization + Deep Neural Network	Accuracy
HawkPhish-DNN (HHO-DNN, 2025)	Harris Hawks Optimization + Deep Neural Network	Precision
HawkPhish-DNN (HHO-DNN, 2025)	Harris Hawks Optimization + Deep Neural Network	Recall
HawkPhish-DNN (HHO-DNN, 2025)	Harris Hawks Optimization + Deep Neural Network	F1-Score
Proposed DeepWAF-XAI	Stacked Bidirectional LSTM + Attention + XAI	Accuracy (Overall)
Proposed DeepWAF-XAI	Stacked Bidirectional LSTM + Attention + XAI	Precision (Weighted)
Proposed DeepWAF-XAI	Stacked Bidirectional LSTM + Attention + XAI	Recall (Weighted)
Proposed DeepWAF-XAI	Stacked Bidirectional LSTM + Attention + XAI	F1-Score (Weighted)
Proposed DeepWAF-XAI	Normal Traffic (LEGAL)	Accuracy
Proposed DeepWAF-XAI	Normal Traffic (LEGAL)	Precision
Proposed DeepWAF-XAI	Normal Traffic (LEGAL)	Recall
Proposed DeepWAF-XAI	Normal Traffic (LEGAL)	F1-Score
Proposed DeepWAF-XAI	SQL Injection (SQL)	Accuracy
Proposed DeepWAF-XAI	SQL Injection (SQL)	Precision
Proposed DeepWAF-XAI	SQL Injection (SQL)	Recall
Proposed DeepWAF-XAI	SQL Injection (SQL)	F1-Score
Proposed DeepWAF-XAI	Cross-Site Scripting (XSS)	Accuracy
Proposed DeepWAF-XAI	Cross-Site Scripting (XSS)	Precision
Proposed DeepWAF-XAI	Cross-Site Scripting (XSS)	Recall

Proposed DeepWAF-XAI	Cross-Site Scripting (XSS)	F1-Score
Proposed DeepWAF-XAI	Path Traversal (SHELL)	Accuracy
Proposed DeepWAF-XAI	Path Traversal (SHELL)	Precision
Proposed DeepWAF-XAI	Path Traversal (SHELL)	Recall
Proposed DeepWAF-XAI	Path Traversal (SHELL)	F1-Score
Proposed DeepWAF-XAI	Phishing URLs (PHISHING)	Accuracy
Proposed DeepWAF-XAI	Phishing URLs (PHISHING)	Precision
Proposed DeepWAF-XAI	Phishing URLs (PHISHING)	Recall
Proposed DeepWAF-XAI	Phishing URLs (PHISHING)	F1-Score

bility attributes.

Reported / Measured Value (%)	Measurement Type
82.40	Literature Benchmark
84.10	Literature Benchmark
79.60	Literature Benchmark
81.80	Literature Benchmark
88.70	Literature Benchmark
89.20	Literature Benchmark
86.50	Literature Benchmark
87.80	Literature Benchmark
91.30	Literature Benchmark
92.00	Literature Benchmark
90.80	Literature Benchmark
91.40	Literature Benchmark
99.82	Literature Benchmark
99.80	Literature Benchmark
99.86	Literature Benchmark

99.83	Literature Benchmark
99.60	Literature Benchmark
99.80	Literature Benchmark
99.40	Literature Benchmark
99.60	Literature Benchmark
92.59	Project-Measured Result
97.14	Project-Measured Result
92.59	Project-Measured Result
94.32	Project-Measured Result
92.40	Project-Measured Result
99.92	Project-Measured Result
92.40	Project-Measured Result
96.01	Project-Measured Result
77.58	Project-Measured Result
14.55	Project-Measured Result
77.58	Project-Measured Result
24.50	Project-Measured Result
95.94	Project-Measured Result
56.24	Project-Measured Result
95.94	Project-Measured Result

70.91	Project-Measured Result
81.25	Project-Measured Result
28.89	Project-Measured Result
81.25	Project-Measured Result
42.62	Project-Measured Result
100.00	Project-Measured Result
91.58	Project-Measured Result
100.00	Project-Measured Result
95.60	Project-Measured Result

Source / Report Reference	Dataset or Evaluation Context
OWASP ModSecurity Core Rule Set Benchmark (CRS v3.3)	Real-world HTTP requests & OWASP test suite
OWASP ModSecurity Core Rule Set Benchmark (CRS v3.3)	Real-world HTTP requests & OWASP test suite
OWASP ModSecurity Core Rule Set Benchmark (CRS v3.3)	Real-world HTTP requests & OWASP test suite
OWASP ModSecurity Core Rule Set Benchmark (CRS v3.3)	Real-world HTTP requests & OWASP test suite
Standard Web Security Benchmark / N-gram Baseline	CSIC 2010 HTTP Dataset
Standard Web Security Benchmark / N-gram Baseline	CSIC 2010 HTTP Dataset
Standard Web Security Benchmark / N-gram Baseline	CSIC 2010 HTTP Dataset
Standard Web Security Benchmark / N-gram Baseline	CSIC 2010 HTTP Dataset
Springer Discover Applied Sciences (2026) 8:23, DOI: 10.1007/s42452-026-06899-7	CSIC 2010 HTTP Dataset (ADASYN oversampled)
Springer Discover Applied Sciences (2026) 8:23, DOI: 10.1007/s42452-026-06899-7	CSIC 2010 HTTP Dataset (ADASYN oversampled)
Springer Discover Applied Sciences (2026) 8:23, DOI: 10.1007/s42452-026-06899-7	CSIC 2010 HTTP Dataset (ADASYN oversampled)
Springer Discover Applied Sciences (2026) 8:23, DOI: 10.1007/s42452-026-06899-7	CSIC 2010 HTTP Dataset (ADASYN oversampled)
Springer (2025), DOI: 10.1007/978-981-97-8899-6_11	Specialized payload dataset (binary SQLi / XSS strings)
Springer (2025), DOI: 10.1007/978-981-97-8899-6_11	Specialized payload dataset (binary SQLi / XSS strings)
Springer (2025), DOI: 10.1007/978-981-97-8899-6_11	Specialized payload dataset (binary SQLi / XSS strings)

Springer (2025), DOI: 10.1007/978-981-97-8899-6_11	Specialized payload dataset (binary SQLi / XSS strings)
Springer Cluster Computing (2025), DOI: 10.1007/s10586-025-05244-6	Phishing URL dataset (PhishTank, Alexa Top 1M)
Springer Cluster Computing (2025), DOI: 10.1007/s10586-025-05244-6	Phishing URL dataset (PhishTank, Alexa Top 1M)
Springer Cluster Computing (2025), DOI: 10.1007/s10586-025-05244-6	Phishing URL dataset (PhishTank, Alexa Top 1M)
Springer Cluster Computing (2025), DOI: 10.1007/s10586-025-05244-6	Phishing URL dataset (PhishTank, Alexa Top 1M)
Project Evaluation Output (results.json)	Stratified held-out test split (n = 19,776 / 197,756 samples)
Project Evaluation Output (results.json)	Stratified held-out test split (n = 19,776 / 197,756 samples)
Project Evaluation Output (results.json)	Stratified held-out test split (n = 19,776 / 197,756 samples)
Project Evaluation Output (results.json)	Stratified held-out test split (n = 19,776 / 197,756 samples)
classification_report_test.txt (support: 18,290)	Stratified held-out test split
classification_report_test.txt (support: 18,290)	Stratified held-out test split
classification_report_test.txt (support: 18,290)	Stratified held-out test split
classification_report_test.txt (support: 18,290)	Stratified held-out test split
classification_report_test.txt (support: 165)	Stratified held-out test split
classification_report_test.txt (support: 165)	Stratified held-out test split
classification_report_test.txt (support: 165)	Stratified held-out test split
classification_report_test.txt (support: 165)	Stratified held-out test split
classification_report_test.txt (support: 789)	Stratified held-out test split
classification_report_test.txt (support: 789)	Stratified held-out test split
classification_report_test.txt (support: 789)	Stratified held-out test split

classification_report_test.txt (support: 789)	Stratified held-out test split
classification_report_test.txt (support: 32)	Stratified held-out test split
classification_report_test.txt (support: 32)	Stratified held-out test split
classification_report_test.txt (support: 32)	Stratified held-out test split
classification_report_test.txt (support: 32)	Stratified held-out test split
classification_report_test.txt (support: 500)	Stratified held-out test split
classification_report_test.txt (support: 500)	Stratified held-out test split
classification_report_test.txt (support: 500)	Stratified held-out test split
classification_report_test.txt (support: 500)	Stratified held-out test split

Notes on Direct Comparability & Scope
Rule-based regex matching; high false-positive rate on complex inputs; lacks representation learning.
Susceptible to regex evasion and URL/unicode encoding tricks.
Frequent false negatives on novel zero-day attack variants.
Baseline harmonic mean across OWASP test partitions.
Shallow statistical classification; lacks sequential contextual memory across variable-length payloads.
Evaluated on tabular n-gram token frequency matrices.
Vulnerable to token-frequency poisoning and payload obfuscation.
Standard machine learning baseline across web application firewall literature.
Evaluated on 3-class traffic; synthetically balanced with ADASYN oversampling; does not evaluate Path Traversal.
Reported benign precision 100.0%, SQLi 76.0%, and XSS 50.0%.
Reported benign recall 100.0%, SQLi 76.0%, and XSS 33.0%.
Weighted harmonic mean across ADASYN synthetic distribution.
Evaluated on isolated binary payload strings; does not evaluate Path Traversal or full multi-class WAF traffic.
Binary classification benchmark on clean isolated tokens.
Near-perfect recall on targeted, isolated binary attack sets.

Payload-level binary classification score.
Specialized solely for URL phishing; does not inspect HTTP payloads, SQL injection, or XSS attacks.
High precision in separating legitimate domain names from phishing URLs.
High phishing intercept rate under HHO hyperparameter search.
Domain-specific phishing benchmark score.
Measured on unaugmented natural imbalanced 5-class web traffic (18,311 / 19,776 correct).
Weighted by class support; Macro Precision is 58.23% due to extreme class imbalance.
Weighted by class support; Macro Recall is 89.43%.
Weighted by class support; Macro F1-Score is 65.93%.
16,900 / 18,290 correctly classified normal requests.
Near-zero false alarms on legitimate business traffic.
Legitimate request passage rate.
High reliability on primary traffic volume.
128 / 165 correctly intercepted SQL injection attempts.
Lower precision due to severe natural class imbalance (165 SQL vs 18,290 Normal).
Solid recall boundary; mitigated further by secondary DB firewall.
Harmonic score under natural production imbalance.
757 / 789 correctly intercepted XSS attack scripts.
Exceeds BALNet published XSS precision (50.0%).
Outperforms BALNet published XSS recall (33.0%) by +62.94%.

Strong balance on script execution payload detection.
26 / 32 correctly intercepted path traversal requests.
Extreme minority class (32 samples out of 19,776).
Effective containment of directory traversal and LFI payloads.
Literature DL models (BALNet, UniEmbed) report N/A on Path Traversal.
500 / 500 correctly intercepted phishing URLs.
High precision distinguishing phishing URLs from legal queries.
Perfect detection recall on held-out phishing test partition.
Competitive with specialized HawkPhish-DNN architecture.